

## Comorbidity of Sexual Problems and Posttraumatic Stress Disorder in Female Crime Victims

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The present study examined the relationship between Posttraumatic Stress Disorder (PTSD) and sexual problems using data from a subset of women ( $n = 391$ ) interviewed about history of crime victimization and *DSM-III*-defined mental health problems (American Psychiatric Association, 1980). The sample was subdivided based on positive ( $n = 228$ ) or negative ( $n = 165$ ) history of sexual problems. These two groups were compared on a number of demographics, crime characteristics, and mental health variables. Variables significantly related to sexual problems included having experienced (a) some form of criminal victimization, (b) completed rape, (c) physical injury during the crime, (d) depression, and (e) PTSD. A logistic regression was conducted with these significant variables. The PTSD variable was entered into the regression model last and was found to contribute a significant amount of the variance after accounting for variance contributed by the other variables. This finding supports the hypothesis that PTSD may act as a moderating variable on the development of sexual problems in women with sexual and nonsexual trauma histories. The findings from the present study are largely consistent with learning theory models of posttraumatic adjustment. Clinical and research implications are discussed.

Over the past decade, hundreds of studies have been reported on the effects of crime and trauma. In several distinct populations of traumatized men and

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women (e.g., war veterans, adult survivors of child sexual assault, and victims of adult rape), it has been reported that sexual problems represent a significant concern (e.g., Becker, Skinner, Abel, & Treacy, 1982; Briere, 1988; Browne & Finkelhor, 1986; Kaplan, 1989; Kilpatrick, Resick, & Veronen, 1981). Despite the apparent robustness of this finding, there is currently no theory that predicts the development of sexual problems in nonsexual assault victims (e.g., veterans), although it has been hypothesized that conditioning processes account for sexual problems in sexual assault victims (e.g., Becker, Skinner, Abel, & Cichon, 1986; Kilpatrick, Veronen, & Resick, 1982).

What is lacking in the present literature is a unifying theory that explains the overall increase in sexual problems in trauma victims, regardless of gender or type of trauma. It is suggested here that trauma victims who develop post-traumatic stress disorder (PTSD) are those individuals at risk for long-term sexual problems; that PTSD acts as a moderating variable in the path between trauma and sexual problems. We focus on PTSD because (a) it is well known that PTSD is a common reaction to highly traumatic events, although by no means does every victim or survivor develop this psychological disorder; and (b) PTSD is associated with all types of trauma (and not, for example, only with sexual victimization). Thus, PTSD appears uniquely qualified to act as a moderating variable in the relationship between victimization and sexual problems. Furthermore, this theory provides testable predictions regarding the development (or not) of sexual problems in trauma victims. Specifically, victims of any trauma who develop PTSD would be at higher risk for sexual problems. Victims of sexual trauma who develop PTSD may be at even higher risk for sexual problems (e.g., due to effects of conditioning or the ability of sexual situations to act as cues for traumatic memories). In addition, it is stated in the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association, 1994) that the ability to feel emotions of any type, "especially those associated with intimacy, tenderness, and sexuality," is often impaired in individuals with PTSD (p. 425). The *DSM* does not differentiate between persons with rape-related PTSD and those with, for example, war-related PTSD. Rather, there is an overall lessening of feelings, including sexual feelings associated with PTSD. Indirect support for these predictions may be found in the literature on the different subgroups of trauma victims reviewed below.

### *Case Studies and Preliminary Research Findings*

Two case studies have been published in which it is suggested that sexual problems occur among individuals with PTSD more frequently than would be expected. In their clinical practices, both Kaplan (1989) and Scrignar (1987) encountered a large percentage of clients who experienced sexual difficulties following onset of PTSD, even when the initial trauma was nonsexual. Kaplan and Scrignar independently concluded that PTSD plays an etiological role in the development of some cases of sexual problems. A similarly high percentage of sexual problems in war veterans with PTSD was reported by

Letourneau and colleagues (Letourneau, Kilpatrick, Resnick, Schewe, & Frueh, 1995). Results of a survey of these men indicated that veterans experienced sexual problems at higher rates than would be expected based on community prevalence rates.

### *Research on Victims of Sexual Abuse*

A number of studies and reviews on the long-term effects of child and adult sexual assault have indicated that sexual assault victims are at increased risk for both sexual problems and anxiety disorders including PTSD<sup>1</sup> (e.g., Becker, Skinner, Abel, & Treacy, 1982; Briere, 1988; Browne & Finkelhor, 1986; Fromuth, 1986; Kilpatrick et al., 1981; Kilpatrick, Veronen, & Resick, 1979; Roesler & McKenzie, 1994). Although none of the studies assessed the comorbidity of PTSD and sexual problems, each lends indirect evidence to the hypothesis that women with PTSD are more likely to have sexual problems.

*Child victims.* Tsai, Feldman-Summers, and Edgar (1979) compared two groups of adult female survivors of child sexual assault (treatment seeking and nontreatment seeking) with a nonvictimized, nonpatient control group. The treatment seeking group reported more severe child sexual assault (e.g., longer duration, more frequent assaults, and more incidents, including attempted vaginal intercourse) than the nontreatment seeking survivor group. With regard to comparisons between all three groups, the treatment seeking survivors indicated more sexual problems (e.g., less likely to experience orgasm during intercourse) than either of the other two groups. The three groups were also compared on MMPI scale scores. The treatment seeking group had higher scores than the remaining groups on seven of the nine subscales, although scores were in the clinical range for only scales 4 (Psychopathic Deviate) and 8 (Schizophrenia). Thus, it appears that all women who experienced child sexual assault are not equally likely to experience sexual problems. Rather, those women who experienced more severe assaults were at increased risk for psychopathology, including sexual problems.

Gorcey, Santiago, and McCall-Perez (1986) examined differences between a nonclinical group of women sexually assaulted in childhood and a nonvictim control group. All assaults involved some form of sexual contact and most would be considered severe sexual assault (e.g., most subjects suffered long-term abuse by a biological or stepfather that involved some form of penetration). Groups were compared on measures of anxiety, fear, and depression. The survivor group was more likely to have elevated depression, higher levels of anxiety, and more fearfulness. The survivor group was also asked a series of open-ended questions on sexual relationships and 85% of these women

<sup>1</sup> Several of the articles reviewed in this section were conducted prior to the formalization of PTSD as a diagnosis (e.g., Becker, Skinner, Abel, & Treacy, 1982; Kilpatrick et al., 1981; Tsai et al., 1979) and, therefore, PTSD could not be assessed. However, authors frequently noted symptoms that are consistent with a diagnosis of PTSD (e.g., nightmares), and this is indicated where applicable.

reported problems. Unfortunately, no group comparison data were reported for this measure.

Briere (1988) compared outpatient clients who had experienced one or more sexual assaults in childhood with a group of nonvictimized outpatient clients. Survivors had generally experienced severe abuse (e.g., 77% experienced penetration and 56% also experienced child physical abuse). The dependent measure included subscales for dissociation, sleep problems, sexual problems, and anger. Survivors scored higher on each of these subscales than the control group.

Fromuth (1986) compared a group of college women sexually assaulted as children with a comparison group of nonassaulted college women. Although assault survivors were more likely to report having engaged in some sexual behaviors (e.g., more noncoital behavior in past month), they were not more likely than the control group to have problems with sexual desire or sexual adjustment. In considering these discrepant findings, it may be that Fromuth reported on a group of survivors less likely to have PTSD than survivors reported in other studies. For example, the abuse suffered by her subjects was less severe than in other studies (e.g., abuse generally involved a single incident of nonpenetrating sexual contact). Thus, Fromuth's findings are not necessarily counter to hypotheses presented in the present paper.

*Adult victims.* Kilpatrick et al. (1979) reported on the psychological consequences of adult rape over four postrape time intervals (6–10 days, 1 month, 3 months, and 6 months postrape). Rape victims were compared with non-victims on measures of fear, anxiety, and mood. Nearly all (25/28) assessment items were elevated for the victim group at the first two time points. However, by 3 and 6 months postrape, only seven items remained elevated. These seven items appeared to be specifically related to the rape experience (e.g., fear of a man's penis, fear of anal intercourse). In a later study reporting on a subset of subjects who completed a final assessment point at 1 year postrape, a similar pattern of results was found, with elevations continuing on a few specific fear items and anxiety subscales (Kilpatrick et al., 1981). Although not designed to assess sexual functioning, the specific fears reported by these women suggest the possibility of sexual problems linked with fear and anxiety symptoms.

Burgess and Holmstrom (1979) assessed several aspects of sexual functioning in a group of adult rape victims who were identified by emergency room personnel and interviewed at several points during the first year postrape and again at 4 to 6 years postrape. They reported that, at the final follow-up assessment point, over two thirds of these women continued to report decreased sexual activity (relative to their own estimates of their sexual activity level prior to the rape). Of the women who had been sexually active postrape, about 25% reported pain during intercourse, and 41% experienced problems with excitement and orgasm. Data regarding other psychological problems were not reported, however the authors noted that about half of these women experienced flashbacks, both during sexual intercourse as well as at other times.

Thus, it is considered possible that some of these women may have had PTSD, in addition to the long-term nature of their sexual problems.

Feldman-Summers, Gordon, and Meagher (1979) obtained retrospective measures on sexual satisfaction and frequency of sexual behaviors from a group of adult rape victims. These measures were compared with those from a control group of nonvictims. Rape victims were asked to complete the assessment measures under three conditions: (a) as if it were 1 week prior to the rape, (b) as if it were 1 week after the rape, and (c) as if it were 2 months after the rape. The victim group reported less satisfaction with sexual activities at 1 week postrape versus 1 week prerape, and satisfaction never returned to prerape levels, although it did increase from 1 week postrape to 2 months postrape. When the victim and nonvictim groups were compared on current sexual functioning, there were no differences in frequency of sexual activities, although victims reported less satisfaction. Thus, rape victims continued to report long-term sexual problems, supporting the presence of a moderating variable.

The results of these studies, and the case studies reported above, are highly suggestive of a relationship between trauma, PTSD, and sexual problems. While conditioning, chaining, and social learning processes have all been offered as explanations for the development of sexual problems following sexual trauma (e.g., Becker, Skinner, Abel, & Cichon, 1984; Becker, Skinner, Abel, & Treacy, 1982; Becker, Skinner, Abel, Howell, & Bruce, 1982), these processes do not account for the development of sexual problems following nonsexual trauma. If we assume that PTSD acts as a moderating variable in the relationship between trauma and sexual problems, it is possible to offer one theory for all subgroups of trauma victims.

In the present study, it was hypothesized that crime victims with PTSD (regardless of type of trauma) would be at greater risk for sexual problems than crime victims who did not have PTSD. Second, it was hypothesized that crime victims with severe sexual victimization (i.e., completed childhood or adult rape) would be at even greater risk for sexual problems than victims of other types of crimes, given the potentially additive effects of classical conditioning and PTSD.

## Method

### *Research Participants*

Respondents who participated in the present study were part of a larger household probability sample of 2,004 women who participated in a parent study. The sampling methodology for the original study is reported in Kilpatrick et al. (1985) and in Kilpatrick et al. (1989). Briefly, households in Charleston, SC, were telephoned based on a random digit dialing method. Households with adult females were requested to participate in the parent study. After the parent study was concluded, 933 study participants were re-contacted to participate in the present study. Of these 933 women, 399 (42.8%)

agreed to participate. Although all 399 women completed the measures, data from 8 participants were excluded from the present analyses due to concerns about their ability to understand the measurement items. Therefore, 391 women were included in the present study. Demographic information for these participants is presented in Table 1.

### *Assessment Instruments*

Two structured interviews were utilized in the present study: the Incident Report Interview (IRI; Kilpatrick, Saunders, Veronen, Best, & Von, 1987) and a modified version of the Diagnostic Interview Schedule (DIS; Robins, Helzer, Croughan, & Ratcliff, 1981).

The IRI was employed to screen for incidents of serious criminal victimization, regardless of whether these incidents had been reported to authorities. The IRI was created for the parent study, and psychometric properties have not been assessed. However, the IRI is composed of highly structured items that are anchored to specific behaviors and that were designed to overcome underreporting biases, especially with regard to sexual crimes. Eight types of crimes were assessed by the IRI, including (1) completed rape (involving penetration of mouth, anus, or vagina without consent and with threat or use of force); (2) attempted rape; (3) completed sexual molestation (involving unwanted sexual contact and force or threat of force but no penetration); (4) attempted sexual molestation; (5) other sexual assault (e.g., voyeurism, exhibitionism); (6) aggravated assault (victim was attacked by an assailant with a weapon or with the intent to kill or seriously injure her); (7) robbery (someone attempted to take something directly from the victim by using force); and (8) burglary (someone tried to break into the home). Crimes were categorized from most severe (1) to least severe (8). When multiple components were involved in the same crime, a woman was classified as having experienced the most severe type of crime disclosed. Demographic data were also collected with the IRI.

The modified DIS was administered by trained clinical psychologists, social workers, and clinical psychology interns. This measure was utilized to determine whether subjects met both lifetime and current criteria for Major Depressive Episode, Agoraphobia, Social Phobia, Simple Phobia, Panic Disorder, and Obsessive Compulsive Disorder, as well as symptoms of psychosis and suicidal behavior and ideation. The DIS is based on the *DSM-III-R* with regard to wording of items and diagnostic decision making. As noted in Kilpatrick et al. (1989), the original DIS was not considered to adequately assess PTSD. Consequently, items were developed to assess PTSD that were based on the *DSM-III* criteria for PTSD. The language and the form for these items were taken directly from the *DSM*. Symptoms did not have to be linked to the trauma (by the respondent) but did have to have at least a 1-week duration. While formal psychometric testing was not conducted with the PTSD subscale, similar prevalence rates for PTSD have been found in more recent studies using in-person interviews and standardized assessment measures

(Resnick, Kilpatrick, Dansky, Saunders, & Best, 1993). Diagnostic reliability was not specifically assessed with this measure.

Sexual problems were also assessed in the modified DIS. Women were asked, "[h]ave you ever had a period of time, lasting two weeks or more, when you had any of the following problems?" Problems assessed were: (1) lack of interest in sex, (2) fear of sex, (3) inability to become sexually excited or aroused, (4) inability to stay sexually excited or aroused, (5) not able to have a climax or orgasm, and (6) painful vaginal intercourse not due to injury or disease. Subjects who responded affirmatively to any one of these six questions composed the "Sexual Problems Positive" group (SPPositive;  $n = 228$ ). Women who denied having had any of these problems composed the "Sexual Problems Negative" group (SPNegative;  $n = 163$ ).

## Results

Tables 1, 2 and 3 include the results from univariate analyses that assessed differences between the SPPositive and SPSNegative groups on (respectively) demographic, crime characteristics, and mental health variables. Due to the high number of analyses, an alpha level of .002 was chosen for significance prior to these tests. This alpha was derived by using the Bonferroni test for

TABLE 1  
COMPARISON OF SPPOSITIVE (SPPos) AND SPNEGATIVE (SPNeg) GROUPS ON DEMOGRAPHICS

| Variable          | SPPos.<br>$n = 228$ | SPNeg.<br>$n = 163$ | Statistic          |
|-------------------|---------------------|---------------------|--------------------|
| Race (% nonwhite) | 25.9%               | 28.8%               | $\chi^2(1) = 00.5$ |
| Employment Status |                     |                     |                    |
| Employed          | 59.2%               | 61.3%               | $\chi^2(2) = 01.8$ |
| Unemployed        | 12.7%               | 16.0%               |                    |
| Housewife         | 28.1%               | 22.7%               |                    |
| Marital Status    |                     |                     |                    |
| Never Married     | 14.3%               | 15.4%               | $\chi^2(2) = 01.0$ |
| Married           | 67.3%               | 62.6%               |                    |
| Div./wid./sep.    | 18.4%               | 22.1%               |                    |
| Education         |                     |                     |                    |
| 11th grade/less   | 13.6%               | 14.1%               | $\chi^2(2) = 00.7$ |
| H. S. graduate    | 45.6%               | 41.7%               |                    |
| College grad.     | 40.8%               | 40.8%               |                    |
| Income            |                     |                     |                    |
| \$0-\$15,000      | 26.2%               | 29.1%               | $\chi^2(3) = 00.7$ |
| \$15,001-\$25,000 | 25.8%               | 28.5%               |                    |
| \$25,001-\$35,000 | 22.6%               | 21.5%               |                    |
| \$35,000 or more  | 25.3%               | 20.9%               |                    |
| Age at Interview  | 41.2                | 42.7                | $F(1,389) = 01.0$  |

TABLE 2  
COMPARISON OF SPPOSITIVE AND SPNEGATIVE GROUPS ON CRIME CHARACTERISTICS

| Variable                          | SPPos.<br><i>n</i> = 228 | SPNeg.<br><i>n</i> = 163 | Statistic            |
|-----------------------------------|--------------------------|--------------------------|----------------------|
| Any Victimization                 | 81.1%                    | 67.5%                    | $\chi^2(1) = 09.6^*$ |
| Injury                            | 24.1%                    | 09.2%                    | $\chi^2(2) = 14.4^*$ |
| Life threat                       | 30.7%                    | 20.9%                    | $\chi^2(1) = 04.7$   |
| Age most recent crime             | 31.1                     | 33.1                     | $F(1,293) = 01.4$    |
| Age at first crime                | 16.7                     | 20.2                     | $F(1,208) = 05.0^a$  |
| Completed rape                    | 29.4%                    | 14.7%                    | $\chi^2(1) = 11.4^*$ |
| Attempted rape                    | 14.9%                    | 10.4%                    | $\chi^2(1) = 01.7$   |
| Completed molestation             | 21.9%                    | 13.5%                    | $\chi^2(1) = 04.5$   |
| Other sexual assault <sup>b</sup> | 08.3%                    | 07.4%                    | $\chi^2(1) = 00.1$   |
| Aggravated assault                | 12.3%                    | 06.1%                    | $\chi^2(1) = 04.1$   |
| Nonphysical crime <sup>c</sup>    | 51.3%                    | 46.0%                    | $\chi^2(1) = 01.1$   |

<sup>a</sup> Only 210 respondents experienced more than one crime.

<sup>b</sup> Includes attempted molestation and sexual assault not involving contact, such as sexual exposure and voyeurism.

<sup>c</sup> Includes robbery, burglary when out of the home, and burglary when at home.

\*  $p < .002$ .

reducing the likelihood of Type 1 error where alpha (.05) is divided by number of analyses (24) (Cohen & Cohen, 1975).

There were no significant differences between groups on any of the demographic variables (Table 1). With regard to the crime characteristics variables (Table 2), subjects in the SPPositive group were more likely to have experienced some form of criminal victimization than women in the SPNegative group (81.1% versus 67.5%,  $p < .002$ ). Subjects in the SPPositive group were also more likely to have experienced an injury during the crime (24.1% versus 9.2%,  $p < .002$ ) and to have experienced a completed rape (29.4% versus 14.7%,  $p < .002$ ). There were no significant differences on the remaining crime characteristics.

With regard to the mental health variables (Table 3), there were two significant findings. Subjects in the SPPositive group were more likely than subjects in the SPNegative group to have experienced depression at some time in their lives (42.5% versus 19.6%,  $p < .002$ ) and to have experienced PTSD (29.4% versus 9.2%,  $p < .002$ ). There were no significant group differences on the remaining mental health variables.

After conducting the univariate analyses, our next step was to determine whether there were significant correlations between the variables identified as significantly related to SPPositivity, and to determine the magnitude and direction of these correlations. Point-biserial correlations were performed between each of the significant variables: Victim, Injury, Rape, Depression, and PTSD. These correlations are reported in Table 4 and each is significant at



TABLE 3  
COMPARISON OF SPPOSITIVE AND SPNEGATIVE GROUPS ON MENTAL HEALTH CHARACTERISTICS

| Variable      | SPPos.<br><i>n</i> = 228 | SPNeg.<br><i>n</i> = 163 | Statistic            |
|---------------|--------------------------|--------------------------|----------------------|
| Agoraphobia   | 07.0%                    | 03.7%                    | $\chi^2(1) = 02.0$   |
| Depression    | 42.5%                    | 19.6%                    | $\chi^2(1) = 22.6^*$ |
| OCD           | 07.5%                    | 01.2%                    | $\chi^2(1) = 08.0$   |
| PTSD          | 29.4%                    | 09.2%                    | $\chi^2(1) = 23.4^*$ |
| Panic Attacks | 05.3%                    | 03.7%                    | $\chi^2(1) = 00.5$   |
| Psychosis     | 03.1%                    | 01.2%                    | $\chi^2(1) = 01.4$   |
| Simple Phobia | 22.8%                    | 14.7%                    | $\chi^2(1) = 04.0$   |
| Social Phobia | 08.8%                    | 06.7%                    | $\chi^2(1) = 00.5$   |

\*  $p < .002$ .

$p < .002$ . In addition, each variable is also significantly correlated with the group variable, indicating positive correlations between these variables and sexual problems.

The third step in these analyses was to determine the unique contribution of each variable to group membership. This was especially important given the overlap demonstrated between variables in the above correlations. A hierarchical logistic regression was employed for this analysis. The crime characteristic variables, Victim, Injury, and Rape, were entered into the model on the first step. Depression was entered in the second step, and PTSD was entered last, to determine whether it contributed to a significant amount of the variance after controlling for all other variables. Odds ratios and 95% confidence intervals are computed from the results of the logistic regression. These ratios indicate the estimated change in the odds of membership in the SPPositive group based on the presence of each of the predictor variables.

In the first step of the equation, the unadjusted odds ratio was 1.51 for the Victim variable, 2.29 for the Injury variable, and 1.64 for the Rape variable. In the second step, the unadjusted odds ratio was 2.52 for Depression. In the

TABLE 4  
ZERO-ORDER CORRELATIONS BETWEEN SIGNIFICANT PREDICTOR VARIABLES  
AND SEXUAL PROBLEMS

|            | Injury | Rape | Depression | PTSD | Sex Prob. |
|------------|--------|------|------------|------|-----------|
| Victim     | .27    | .31  | .16        | .29  | .16       |
| Injury     |        | .37  | .21        | .38  | .19       |
| Rape       |        |      | .19        | .49  | .17       |
| Depression |        |      |            | .32  | .24       |
| PTSD       |        |      |            |      | .24       |

Note. Two-tailed significance of at least  $p < .01$  occurred for all correlations.

TABLE 5  
LOGISTIC REGRESSION TESTING THE RELATIONSHIP BETWEEN VICTIM STATUS, INJURY,  
RAPE, DEPRESSION, AND PTSD FOR SPPOSITIVE GROUP

| Interval   | Odds Ratio | Confidence Interval |
|------------|------------|---------------------|
| Step 1     |            |                     |
| Victim     | 01.3       | 0.77-2.16           |
| Injury     | 01.7       | 0.90-3.56           |
| Rape       | 01.2       | 0.62-3.10           |
| Step 2     |            |                     |
| Depression | 02.3*      | 1.38-3.67           |
| Step 3     |            |                     |
| PTSD       | 02.3*      | 1.13-4.66           |

\*  $p < .05$ .

third and final step, the odds ratios are adjusted for the inclusion of all variables. Thus, those that are significant contribute to a significant amount of the variance of group membership, controlling for each of the other variables entered. These results are presented in Table 5. In this third step, only depression and PTSD remained significant ( $p < .05$ ) after accounting for the effects of the other variables. The adjusted odds ratios were 2.26 for Depression and 2.30 for PTSD. This indicates that women who have experienced depression at some point in their lives are at two and one thirds greater risk for being in the SPPositive group than women who never experienced depression. Similarly, women who experienced PTSD at some point in their lives are at two and one thirds greater risk for being in the SPPositive group than women who never experienced PTSD. Once these two variables are controlled for, victimization, injury, and rape histories no longer contribute a significant amount to the variance of group membership in this model. A 95% confidence interval also was calculated for each adjusted odds ratio (see Table 5).

After these analyses were completed, we assessed whether this same pattern of results would occur utilizing a more stringent assessment of sexual problems. Specifically, a subgroup was formed with women who endorsed three or more of the six sexual problem items from the DIS. The items most frequently endorsed by this group (High SPPositive,  $n = 92$ ) were: (a) "Lack of interest in sex" (91%), (b) "Not able to climax or orgasm" (84%), and (c) "Not able to become sexually aroused" (83%).

Univariate comparisons between the High SPPositive group and the SPNegative group ( $n = 196$ ) were conducted with the same variables as those used in analyses comparing the SPPositive and SPNegative groups. Subjects who endorsed only one or two sexual problems were not included in the present set of analyses.

There were no significant differences between the High SPPositive group and the SPNegative group on demographic variables. Regarding the crime

TABLE 6  
LOGISTIC REGRESSION TESTING THE RELATIONSHIP BETWEEN INJURY, RAPE, DEPRESSION,  
O-C DISORDER, AND PTSD FOR THE HIGH SPPOSITIVE SUBGROUP

| Interval   | Odds Ratio | Confidence Interval |
|------------|------------|---------------------|
| Step 1     |            |                     |
| Injury     | 2.66*      | 1.22-5.81           |
| Rape       | 1.24       | 0.57-2.69           |
| Step 2     |            |                     |
| Depression | 2.03*      | 1.05-3.86           |
| O-C        | 5.88*      | 1.08-31.82          |
| Step 3     |            |                     |
| PTSD       | 2.77*      | 1.20-6.42           |

\*  $p < .05$ .

characteristic variables, subjects in the High SPPositive subgroup were significantly more likely than subjects in the SPNegative group to have experienced rape (38.0% versus 14.7%,  $p < .001$ ) and injury during the crime (33.7% versus 9.2%,  $p < .001$ ). Regarding the mental health variables, subjects in the High SPPositive group were significantly more likely to have experienced depression (47.8% versus 19.6%,  $p < .001$ ), Obsessive Compulsive Disorder (10.9% versus 1.2%,  $p < .001$ ), and PTSD (39.1% versus 9.2%,  $p < .001$ ).

A logistic regression employing these five significant variables was conducted, with Injury and Rape entered in the first step. The unadjusted odds ratios for these two variables were 3.73 and 2.52, respectively. On the second step, Depression and Obsessive Compulsive Disorder were entered. This resulted in unadjusted odds ratios of 2.53 for Depression and 6.36 for Obsessive Compulsive Disorder. On the third step, PTSD received an unadjusted odds ratio of 2.77. The final model for this logistic regression is presented in Table 6. As can be seen in Table 6, Injury, Depression, Obsessive Compulsive Disorder, and PTSD continued to contribute a significant amount of the variance in the final model. This indicates that women with these crime and mental health experiences are significantly more likely also to have several sexual problems.

## Discussion

It has been argued in the present study that PTSD acts as a moderating variable by impacting on the likelihood that trauma victims will develop long-term sexual problems. Indirect evidence for this theory is provided by case studies, preliminary research findings, and literature on both child and adult sexual victimization. The present study attempted to further delineate the potential relationship between PTSD and sexual problems. Two specific hypotheses were tested. First, it was hypothesized that women who had experienced

PTSD at some point in their lives would be at increased risk for sexual problems than women without a history of PTSD. This hypothesis was supported by univariate analyses that demonstrated a positive relationship between PTSD and sexual problems in two groups (SPPositive and High SPPositive). Specifically, nearly 30% of the SPPositive group and nearly 40% of the High SPPositive group had also experienced some form of PTSD, whereas less than 10% of the SPNegative group had experienced PTSD. Depression was the only other mental health variable associated with membership in the SPPositive group. Although not a specific hypothesis of the present study, it has been well documented that depression is a risk factor for sexual problems (Channon & Ballinger, 1986; Schover, 1984; Sultan & Chambless, 1988). However, PTSD continued to contribute a significant amount to the variance of group membership after accounting for the effects of depression. This finding suggests something unique about PTSD as associated with sexual problems. It may be that depression suppresses several behaviors (e.g., sexual functioning, eating, sleeping) whereas PTSD has a more specific effect on sexual functioning, as is suggested in the *DSM-IV*. The frequent comorbidity of depression and PTSD may make it difficult to differentiate the effect of each disorder on sexual functioning, and both appear to contribute an equal amount to the variance of the computed regressions. Future research should include a focus on the delineation of the relative contributions of depression and PTSD to sexual problems, as well as the unique contributions of specific PTSD symptoms.

In the logistic regression conducted with the High SPPositive group, PTSD, Depression, and Obsessive Compulsive Disorder remained significant contributors to the variance. It was not predicted that Obsessive Compulsive Disorder would be associated with greater sexual problems. Previous studies rarely assess this variable and, when Obsessive Compulsive Disorder is assessed, the findings are mixed (e.g., Greenwald, Leitenberg, Cado, & Tarran, 1990; Kilpatrick et al., 1981). There is a clear need for replication before conclusions can be made regarding this specific finding.

It was also hypothesized that women who experienced some form of penetration during a sexual assault (i.e., rape) would be at greater risk for sexual problems than victims of other types of crimes. This hypothesis was supported in the univariate analyses that found rape to be positively associated with the SPPositive and High SPPositive groups. Nearly 30% of the SPPositive group and nearly 40% of the High SPPositive group had experienced a completed rape, compared with about 15% of the SPNegative group. There were no significant differences between groups with respect to history of other crime types, although women in the SPPositive group were more likely to have some type of victimization history than were women in the SPNegative group.

Interestingly, women who reported having experienced other types of sexual crimes (but not rape) were not more likely to be in the SPPositive or High SPPositive groups. This finding runs counter to the findings of several studies in which women who had been molested as children were more likely to have sexual problems later in life than women who had not been molested (Jehu,

1989; Moeller, Bachmann, & Moeller, 1993; Gorcey et al., 1986). It is important to note that a trend was observed relating childhood molestation and sexual problems. As can be seen in Table 2, the chi-square for completed molestation was 4.5 ( $p = .03$ ). Alternatively, it may be that the present study employed more specific criteria for grouping women based on type of sex crime. We separated rape from nonpenetrating molestation and noncontact sexual abuse, and these groups may have overlapped somewhat in other studies. The present study also employed a more representative sample of women than previous studies. Nevertheless, molestation and other sexual crimes should continue to be considered in future research so as to determine whether our findings generalize across groups.

Women who had experienced physical injury during their victimization were also more likely to be in the SPPositive and High SPPositive groups. This relationship was not predicted. However, sustaining an injury during a crime has been shown to be related to the development of PTSD (Kilpatrick et al., 1989). Thus, as with rape, it may be that injury is a predictor of sexual problems to the extent that it is associated with PTSD. This relationship will need to be addressed in future research.

Previous research has also demonstrated that the subjective perception of life threat during a crime is associated with the development of PTSD (Kilpatrick et al., 1989). However, this variable was not significantly associated with the development of sexual problems in the present study. Again, there was a trend relating the perception of life threat to PTSD. The chi-square analysis employing Life Threat was 4.72 ( $p = .03$ ). Thus, it is recommended that researchers continue to examine this variable.

The results of the present study are consistent with a learning model of sexual problems, such as those proposed by Kilpatrick et al. (1982) and Becker et al. (1986). Specifically, it may be that women who experience a sexual crime involving physical force, sexual penetration, and pain, are immediately conditioned to respond with negative feelings (e.g., fear, anger) to the particular crime stimuli. Maintenance and generalization of these negative feelings may occur via PTSD, where positive feelings are often suppressed, and where avoidance of cues is maintained. More specifically, with PTSD, victims would be less likely to engage in sexual activities during which extinction would be possible. Furthermore, women who do engage in consensual sexual activities may continue to experience sexual problems if they have PTSD due to the effects of emotional numbing. Without PTSD, aversive conditioned responses may more readily be extinguished due to more frequent exposure to sexual cues and due to the ability of the individual to experience positive emotions in conjunction with these exposures.

The present study was limited in that it was based on retrospective information and the items used to gather information about sexual problems were underdeveloped. A brief measure of sexual problems, such as was used in the present study, cannot substitute for a thorough assessment for sexual dysfunction. It is strongly recommended that researchers and clinicians working

with trauma victims employ more standard assessment measures of sexual dysfunction. Researchers and clinicians should also make an effort to determine whether the sexual symptoms are related to the trauma, or whether they represent a more long-term pattern of sexual behavior. Likewise, it is recommended that researchers and clinicians working with clients who have sexual problems employ a standard assessment measure of PTSD.

Given the findings of the present study, it is suggested that future research focus on determining whether individuals with PTSD-related sexual problems have different needs from other clients with sexual problems. It is possible that sexual problems diminish over the course of PTSD treatment. However, it may also be that clients with PTSD and sexual problems require a component of therapy that specifically targets those problems. Becker et al. (1984) reported on the treatment of sexual problems in a group of women who had been sexually assaulted. Their behaviorally specific treatment package included traditional treatment components (e.g., sensate focus exercises and masturbatory training) as well as exercises designed to expose women to "fear-inducing sexual situations, behaviors, and interactions" (p. 98). This combination of exposure and traditional components was successful in ameliorating sexual problems in most of their clients.

Given the potential negative impact of sexual problems on relationships and self-esteem, it is hoped that, as both clinicians and researchers, we will continue to focus on this issue to better understand and meet the needs of our clients.

## References

- American Psychiatric Association (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: Author.
- American Psychiatric Association (1980). *Diagnostic and statistical manual of mental disorders* (3rd ed.). Washington, DC: Author.
- Becker, J. V., Skinner, L. J., Abel, G. G., & Cichon, J. (1984). Time limited therapy with sexually dysfunctional sexually assaulted women. *Journal of Social Work and Human Sexuality*, 3, 97-115.
- Becker, J. V., Skinner, L. J., Abel, G. G., & Cichon, J. (1986). Level of postassault sexual functioning in rape and incest victims. *Archives of Sexual Behavior*, 15, 37-49.
- Becker, J. V., Skinner, L. J., Abel, G. G., Howell, J., & Bruce, K. (1982). The effects of sexual assault on rape and attempted rape victims. *Victimology: An International Journal*, 7, 106-113.
- Becker, J. V., Skinner, L. J., Abel, G. G., & Treacy, E. C. (1982). Incidence and types of sexual dysfunctions in rape and incest victims. *Journal of Sex and Marital Therapy*, 8, 65-74.
- Briere, J. (1988). The long-term clinical correlates of childhood sexual victimization. *Annals of the New York Academy of Sciences*, 528, 327-334.
- Browne, A., & Finkelhor, D. (1986). Impact of child sexual abuse: A review of the research. *Psychological Bulletin*, 99, 555-584.
- Burgess, A. W., & Holmstrom, L. L. (1979). Rape: Sexual disruption and recovery. *American Journal of Orthopsychiatry*, 49, 648-657.
- Channon, L. D., & Ballinger, S. E. (1986). Some aspects of sexuality and vaginal symptoms during menopause and their relations to anxiety and depression. *British Journal of Medical Psychology*, 59, 173-180.

- Cohen, J., & Cohen, P. (1975). *Applied multiple regression/correlation analysis for the behavioral sciences*. Hillsdale, NJ: John Wiley & Sons.
- Feldman-Summers, S., Gordon, P. E., & Meagher, J. R. (1979). The impact of rape on sexual functioning. *Journal of Abnormal Psychology*, 88, 101-105.
- Fromuth, M. E. (1986). The relationship of childhood sexual abuse with later psychological and sexual adjustment in a sample of college women. *Child Abuse & Neglect*, 10, 5-15.
- Gorcey, M., Santiago, J. M., & McCall-Perez, F. (1986). Psychological consequences for women sexually abused in childhood. *Social Psychiatry*, 21, 129-133.
- Greenwald, E., Leitenberg, H., Cado, S., & Tarran, M. J. (1990). Childhood sexual abuse: Long-term effects on psychological and sexual functioning in a nonclinical and nonstudent sample of adult women. *Child Abuse & Neglect*, 14, 503-513.
- Jehu, D. (1989). Mood disturbances among women clients sexually abused in childhood: Prevalence, etiology, treatment. *Journal of Interpersonal Violence*, 4, 164-184.
- Kaplan, P. (1989). Post-traumatic stress syndrome and sexual dysfunction. *Journal of Sex and Marital Therapy*, 15, 74-77.
- Kilpatrick, D. G., Best, C. L., Veronen, J. J., Amick, A. E., Villeponteaux, L. A., & Ruff, G. A. (1985). Mental health correlates of criminal victimization: A random community survey. *Journal of Consulting and Clinical Psychology*, 53, 866-873.
- Kilpatrick, D. G., Resick, P. A., & Veronen, L. J. (1981). Effects of a rape experience: A longitudinal study. *Journal of Social Issues*, 37, 105-122.
- Kilpatrick, D. G., Saunders, B. E., Amick-McMullan, A., Best, C. L., Veronen, L. J., & Resnick, H. S. (1989). Victim and crime factors associated with the development of crime-related post-traumatic stress disorder. *Behavior Therapy*, 20, 199-214.
- Kilpatrick, D. G., Saunders, B. E., Veronen, L. J., Best, C. L., & Von, J. M. (1987). Criminal victimization: Lifetime prevalence reporting to police, and psychological impact. *Crime and Delinquency*, 33, 479-489.
- Kilpatrick, D. G., Veronen, L. J., & Resick, P. A. (1979). The aftermath of rape: Recent empirical findings. *American Journal of Orthopsychiatry*, 49, 658-669.
- Kilpatrick, D. G., Veronen, L. J., & Resick, P. A. (1982). Psychological sequelae to rape. In D. M. Dolesy, R. L. Meredith, & A. R. Ciminero (Eds.), *Behavioral medicine: Assessment and treatment strategies* (pp. 473-497). New York: Plenum.
- Letourneau, E. J., Kilpatrick, D., Resnick, H., Schewe, P., & Frueh, B. C. (1995, November). Comorbidity of sexual problems and PTSD. In D. Riggs (Chair), *Intimacy and relationships in the wake of trauma*. Symposium conducted at the Annual Association for Advancement of Behavior Therapy Meeting, Washington, DC.
- Moeller, T. P., Bachmann, G., & Moeller, J. R. (1993). The combined effects of physical, sexual, and emotional abuse during childhood: Long-term health consequences for women. *Child Abuse & Neglect*, 17, 623-640.
- Resnick, H. S., Kilpatrick, D. G., Dansky, B. S., Saunders, B. E., & Best, C. L. (1993). Prevalence of civilian trauma and posttraumatic stress disorder in a representative national sample of women. *Journal of Consulting and Clinical Psychology*, 61, 984-991.
- Robins, L. N., Helzer, J. D., Croughan, J., & Ratcliff, K. S. (1981). The National Institute of Mental Health Diagnostic Interview Schedule: Its history, characteristics, and validity. *Archives of General Psychiatry*, 38, 381-389.
- Roesler, T. A. & McKenzie, N. (1994). Effects of childhood trauma on psychological functioning in adults sexually abused as children. *The Journal of Nervous and Mental Disease*, 182, 145-150.
- Schover, L. R. (1984). Sexual problems in chronic illness. In S. R. Leiblum & R. C. Rosen (Eds.), *Principles and practice of sex therapy: Update for the 1990's* (2nd ed.), (pp. 319-351). New York: Guilford Press.

- Scrignar, C. B. (1987). Posttraumatic stress disorder and sexual dysfunction. *Medical Aspects of Human Sexuality*, 21, 102-112.
- Sultan, F. E., & Chambless, D. L. (1988). Sexual functioning. In E. A. Blechman & K. D. Brownell (Eds.), *Handbook of behavioral medicine for women: Pergamon general psychology series* (Vol. 149, pp. 92-102). Oxford, England: Pergamon Press.
- Tsai, M., Feldman-Summers, S., & Edgar, M. (1979). Childhood molestation: Variables related to differential impacts on psychosexual functioning in adult women. *Journal of Abnormal Psychology*, 88, 407-417.

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